

Human Review: Pringsheim-singular spaceability question

Original automatic proof: `solution_packet.pdf`

Checked date: 13 June 2026

Status: Manually checked.

Verdict: Not a solution to the original question as stated in Bernal-Gonzalez–Bonilla–Lopez-Salazar–Seoane-Sepulveda.

Human Comments

The proposed construction appears to prove a weaker natural-boundary statement, but it does not prove the Pringsheim-singularity condition required in the original question.

The source packet interprets “Pringsheim singular at every boundary point” as meaning that the holomorphic function has no analytic continuation through any point of $\mathbb{T}$. Under this interpretation, Hadamard’s gap theorem is a natural tool: the constructed functions have lacunary Taylor support and radius of convergence exactly 1, so the unit circle is a natural boundary.

However, the original paper

Bernal-Gonzalez–Bonilla–Lopez-Salazar–Seoane-Sepulveda,

Nowhere holderian functions and Pringsheim singular functions in the disc algebra, defines Pringsheim singularity for the boundary function in a stronger real-variable sense. If

$$f_0(t) = f(e^{it}),$$

then $f|_{\mathbb{T}}$ is Pringsheim-singular at e^{it_0} when the Taylor series of f_0 at t_0 has radius of convergence zero. Equivalently,

$$\limsup_{r \rightarrow \infty} \left| \frac{f_0^{(r)}(t_0)}{r!} \right|^{1/r} = +\infty.$$

This is stronger than merely saying that f has no holomorphic continuation through e^{it_0} .

Where the Proof Falls Short

The packet proves that every nonzero element of the constructed closed span has $\mathbb{T}$ as a natural boundary. This rules out analytic continuation through any boundary point, and plausibly gives a nowhere-real-analytic boundary function. But it does not establish the derivative-growth condition above.

For a boundary function

$$f_0(t) = \sum_{n \in \Lambda} c_n e^{int},$$

one has

$$f_0^{(r)}(t_0) = \sum_{n \in \Lambda} (in)^r c_n e^{int_0}.$$

To prove Pringsheim singularity in the sense of the original paper, one would need pointwise lower bounds on these derivative sums at every t_0 , strong enough to force

$$\left| f_0^{(r)}(t_0)/r! \right|^{1/r}$$

to be unbounded.

The coefficient choice $c_n = e^{-\sqrt{n}}$ is suggestive: a single term of the form $n^r e^{-\sqrt{n}}$ can be very large compared with $r!$ for appropriate n . But the proof does not show that such a term dominates the whole derivative sum, nor does it rule out cancellations. This is especially important for general nonzero elements of the closed span, where several lacunary blocks can contribute simultaneously with complex coefficients.

Review Outcome

This packet should be rejected as a solution to the original Pringsheim spaceability question. It should not be archived as a correct full answer to the question in [6].

It may still be useful as a proof of a weaker statement: spaceability of smooth disk-algebra functions whose unit circle is a natural boundary, or possibly whose boundary restrictions are nowhere real analytic. But the argument does not prove that the boundary Taylor radius is zero at every point, which is the condition required for Pringsheim singularity in the original reference.